

COM3031-Probabilistic Machine Learning Workshop:

Maximising Likelihood Estimation

Overview

This workshop consists of three questions. In the first question, I provide a demonstration of maximising likelihood to estimate the mean and variance of the normal distribution. In the second question, you will estimate the parameter of the given distribution by maximising the given likelihood function. In the third question, you will first find the appropriate distribution and maximise the likelihood to estimate the parameters. I will publish the solution in python next week. You may want to go through the lecture slides on maximising likelihood.

Maximising likelihood estimation

1. Assume a data $\mathbf{y} = (y_1, \dots, y_n)^T$ is normally distributed. Write a script to get the mean and variance by maximising the likelihood.
2. Von Bortkiewicz counted the number of Prussian cavalrymen who were killed to death by a horse each year during 1875-1894. The data has been provided to you on ELE. This is the count data and a Poisson distribution can be used to model the data.

$$\mathbf{y} \sim \text{Poisson}(\lambda) \quad \lambda > 0$$

$$p(\mathbf{y}|\lambda) \propto \prod_{i=1}^n \lambda^{y_i} e^{-\lambda}$$

where λ is the parameter and also describes the mean in the distribution. Write a script to estimate the parameter λ by maximising the likelihood.

3. Probability of getting heads in a fixed number of trials: Assume you flip a coin N times. What is the appropriate distribution to describe the data set? Write a script to maximise the likelihood and estimate the probability of getting heads. You can select arbitrary values for N and number of heads.